

Leave Management System - API Endpoints Documentation

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Base URL: http://localhost (via Traefik)

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Overview

The Leave Management System is a microservices-based platform composed of three independent services:

Service	Port	Description
Auth Service	8001	User authentication and JWT token generation
Employee Service	8002	Employee information and leave balance management
Leave Service	8003	Leave request management and approval workflows

All services are accessed through the Traefik reverse proxy on port 8000.

Authentication

Authentication Flow

1. User calls `/auth/login` endpoint with credentials
2. Server validates credentials and returns JWT token
3. Client includes token in `Authorization` header for subsequent requests
4. Token format: `Authorization: Bearer <access_token>`

Token Details

- **Type:** JWT (JSON Web Token)
- **Expiration:** 24 hours (86400 seconds)
- **Algorithm:** HS256
- **Claims:** Includes `user_id` and `role`

Required Headers

All authenticated endpoints require:

```
Authorization: Bearer <jwt_token>  
Content-Type: application/json
```

Auth Service

POST /auth/login

Authenticate user and obtain JWT token for subsequent requests.

Endpoint: POST `http://localhost/auth/login`

Access: Public (No authentication required)

Request Headers:

```
Content-Type: application/json
```

Request Body:

```
{
  "username": "john_doe",
  "password": "secure_password123"
}
```

Request Body Schema:

Field	Type	Required	Description
username	string	Yes	Username of the user
password	string	Yes	Password of the user

Success Response:

```
HTTP/1.1 200 OK
Content-Type: application/json

{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VyX2lkIjoiam9ob19kb2UiLCJyb2x1Ijo:
  "token_type": "bearer",
  "expires_in": 86400,
  "role": "EMPLOYEE"
}
```

Response Schema:

Field	Type	Description
access_token	string	JWT token for authenticated requests
token_type	string	Token type (always "bearer")
expires_in	integer	Token expiration time in seconds
role	string	User role (EMPLOYEE, MANAGER, or ADMIN)

Error Responses:

HTTP/1.1 401 Unauthorized
Content-Type: application/json

```
{  
  "detail": "Invalid username or password"  
}
```

cURL Example:

```
curl -X POST "http://localhost/auth/login" \  
-H "Content-Type: application/json" \  
-d '{  
  "username": "john_doe",  
  "password": "secure_password123"  
'
```

Employee Service

GET /employees/me

Retrieve the current authenticated user's employee information.

Endpoint: GET http://localhost/employees/me

Access: Authenticated (All roles)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Query Parameters: None

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "id": 1,
  "username": "john_doe",
  "full_name": "John Doe",
  "role": "EMPLOYEE",
  "manager_id": 2
}
```

Response Schema:

Field	Type	Description
id	integer	Unique employee ID
username	string	Username of the employee
full_name	string	Full name of the employee
role	string	Employee role (EMPLOYEE or MANAGER)
manager_id	integer	ID of the reporting manager (null if none)

Error Responses:

HTTP/1.1 401 Unauthorized
Content-Type: application/json

```
{
  "detail": "Not authenticated"
}
```

cURL Example:

```
curl -X GET "http://localhost/employees/me" \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \
-H "Content-Type: application/json"
```

GET /employees/leave-balances

Retrieve leave balances for all leave types for the current user.

Endpoint: GET `http://localhost/employees/leave-balances`

Access: Authenticated (All roles)

Request Headers:

Authorization: Bearer <jwt_token>

Content-Type: application/json

Query Parameters: None

Success Response:

HTTP/1.1 200 OK

Content-Type: application/json

```
[
  {
    "id": 101,
    "employee_id": 1,
    "leave_type": "CASUAL",
    "allocated": 12,
    "used": 3,
    "remaining": 9
  },
  {
    "id": 102,
    "employee_id": 1,
    "leave_type": "SICK",
    "allocated": 10,
    "used": 1,
    "remaining": 9
  },
  {
    "id": 103,
    "employee_id": 1,
    "leave_type": "EARNED",
    "allocated": 5,
    "used": 0,
    "remaining": 5
  },
  {
    "id": 104,
    "employee_id": 1,
    "leave_type": "UNPAID",
    "allocated": 0,
    "used": 0,
    "remaining": 0
  }
]
```

Response Schema (Array):

Field	Type	Description
id	integer	Unique leave balance ID
employee_id	integer	Employee ID

Field	Type	Description
leave_type	string	Type of leave (CASUAL, SICK, EARNED, UNPAID)
allocated	integer	Total allocated days
used	integer	Days already used
remaining	integer	Remaining balance

Error Responses:

HTTP/1.1 401 Unauthorized
Content-Type: application/json

```
{  
  "detail": "Not authenticated"  
}
```

cURL Example:

```
curl -X GET "http://localhost/employees/leave-balances" \  
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
-H "Content-Type: application/json"
```

GET /employees/{username}/leave-balance/

Retrieve specific leave balance for a particular employee and leave type.

Endpoint: GET http://localhost/employees/{username}/leave-balance/{leave_type}

Access: Authenticated (Employee can view their own, Managers can view their team)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Path Parameters:

Parameter	Type	Required	Description
username	string	Yes	Username of employee
leave_type	string	Yes	Leave type (CASUAL, SICK, EARNED, UNPAID)

Query Parameters: None

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "leave_type": "CASUAL",
  "allocated": 12,
  "used": 3,
  "remaining": 9
}
```

Response Schema:

Field	Type	Description
leave_type	string	Type of leave
allocated	integer	Total allocated days
used	integer	Days already used
remaining	integer	Remaining balance

Error Responses:

HTTP/1.1 404 Not Found
Content-Type: application/json

```
{
  "detail": "Employee not found"
}
```

HTTP/1.1 403 Forbidden
Content-Type: application/json

```
{  
  "detail": "Access denied"  
}
```

cURL Example:

```
curl -X GET "http://localhost/employees/john_doe/leave-balance/CASUAL" \  
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
-H "Content-Type: application/json"
```

PUT /employees/{username}/deduct-balance

Deduct leave balance for an employee (used internally when approving leaves).

Endpoint: PUT http://localhost/employees/{username}/deduct-balance

Access: Authenticated (Service-to-service, Managers, or Admin)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Path Parameters:

Parameter	Type	Required	Description
username	string	Yes	Username of employee

Request Body:

```
{
  "leave_type": "CASUAL",
  "days": 3
}
```

Request Body Schema:

Field	Type	Required	Description
leave_type	string	Yes	Type of leave (CASUAL, SICK, EARNED, UNPAID)
days	integer	Yes	Number of days to deduct (must be > 0)

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "message": "Leave balance updated",
  "remaining": 6
}
```

Response Schema:

Field	Type	Description
message	string	Success message
remaining	integer	Updated remaining balance

Error Responses:

HTTP/1.1 400 Bad Request
Content-Type: application/json

```
{
  "detail": "Insufficient leave balance"
}
```

HTTP/1.1 404 Not Found

Content-Type: application/json

```
{
  "detail": "Employee not found"
}
```

cURL Example:

```
curl -X PUT "http://localhost/employees/john_doe/deduct-balance" \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \
-H "Content-Type: application/json" \
-d '{
  "leave_type": "CASUAL",
  "days": 3
}'
```

GET /employees/{manager_username}/team

Retrieve list of employees reporting to a manager.

Endpoint: GET `http://localhost/employees/team/{manager_username}`

Access: Authenticated (Admin or the manager themselves)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Path Parameters:

Parameter	Type	Required	Description
manager_username	string	Yes	Username of the manager

Query Parameters: None

Success Response:

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
[
  "john_doe",
  "alice_johnson",
  "bob_williams"
]
```

Response Schema: Array of usernames (strings)

Error Responses:

```
HTTP/1.1 403 Forbidden
Content-Type: application/json
```

```
{
  "detail": "Not allowed to view this team"
}
```

```
HTTP/1.1 404 Not Found
Content-Type: application/json
```

```
{
  "detail": "Manager not found"
}
```

cURL Example:

```
curl -X GET "http://localhost/employees/team/jane_smith" \
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \
  -H "Content-Type: application/json"
```

GET /employees/{username}/manager

Retrieve the manager information for a specific employee.

Endpoint: GET http://localhost/employees/{username}/manager

Access: Authenticated (Employee can view their own, Managers can view their team)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Path Parameters:

Parameter	Type	Required	Description
username	string	Yes	Username of the employee

Query Parameters: None

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "employee": "john_doe",
  "manager": "jane_smith"
}
```

Response Schema:

Field	Type	Description
employee	string	Username of the employee
manager	string	Username of the manager (null if no manager)

Error Responses:

HTTP/1.1 404 Not Found

Content-Type: application/json

```
{  
  "detail": "Employee not found"  
}
```

cURL Example:

```
curl -X GET "http://localhost/employees/john_doe/manager" \  
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
-H "Content-Type: application/json"
```

Leave Service

POST /leaves/apply

Submit a new leave request.

Endpoint: POST http://localhost/leaves/apply

Access: Authenticated (All roles)

Request Headers:

Authorization: Bearer <jwt_token>

Content-Type: application/json

Request Body:

```
{
  "leave_type": "CASUAL",
  "start_date": "2026-06-15",
  "end_date": "2026-06-17",
  "number_of_days": 3,
  "reason": "Personal business"
}
```

Request Body Schema:

Field	Type	Required	Description	Constraints
leave_type	string	Yes	Type of leave	CASUAL, SICK, EARNED, UNPAID
start_date	date	Yes	Leave start date	YYYY-MM-DD format, not in past
end_date	date	Yes	Leave end date	YYYY-MM-DD format, >= start_date
number_of_days	integer	Yes	Number of days	Must match date range, > 0
reason	string	Yes	Reason for leave	1-500 characters

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "message": "Leave applied successfully",
  "leave_id": 42,
  "status": "PENDING"
}
```

Response Schema:

Field	Type	Description
message	string	Success message

Field	Type	Description
leave_id	integer	Unique ID of the leave request
status	string	Current status (PENDING)

Error Responses:

HTTP/1.1 400 Bad Request
Content-Type: application/json

```
{
  "detail": "Insufficient leave balance. Remaining: 2"
}
```

HTTP/1.1 409 Conflict
Content-Type: application/json

```
{
  "detail": "Overlapping leave request exists"
}
```

HTTP/1.1 503 Service Unavailable
Content-Type: application/json

```
{
  "detail": "Employee Service temporarily unavailable"
}
```

Validation Examples:

Invalid - Dates in past:

```
{
  "leave_type": "CASUAL",
  "start_date": "2026-05-15",
  "end_date": "2026-05-17",
  "number_of_days": 3,
  "reason": "Personal business"
}
```

Response: {"detail": "Cannot apply for past dates"}

Invalid - End date before start date:

```
{
  "leave_type": "CASUAL",
  "start_date": "2026-07-20",
  "end_date": "2026-07-15",
  "number_of_days": 5,
  "reason": "Personal business"
}
```

Response: {"detail": "End date cannot be before start date"}

Invalid - Number of days mismatch:

```
{
  "leave_type": "CASUAL",
  "start_date": "2026-06-15",
  "end_date": "2026-06-17",
  "number_of_days": 5,
  "reason": "Personal business"
}
```

Response: {"detail": "number_of_days must be 3 for the selected date range"}

cURL Example:

```
curl -X POST "http://localhost/leaves/apply" \
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \
  -H "Content-Type: application/json" \
  -d '{
    "leave_type": "CASUAL",
    "start_date": "2026-06-15",
    "end_date": "2026-06-17",
    "number_of_days": 3,
    "reason": "Personal business"
  }'
```

GET /leaves/history

Retrieve leave request history for the current user with pagination and filtering.

Endpoint: GET http://localhost/leaves/history

Access: Authenticated (All roles)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Query Parameters:

Parameter	Type	Default	Description
skip	integer	0	Number of records to skip (pagination)
limit	integer	10	Number of records to return
status	string	None	Filter by status (PENDING, APPROVED, REJECTED, CANCELLED)

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```

{
  "items": [
    {
      "id": 42,
      "employee_username": "john_doe",
      "leave_type": "CASUAL",
      "start_date": "2026-06-15",
      "end_date": "2026-06-17",
      "number_of_days": 3,
      "reason": "Personal business",
      "status": "APPROVED",
      "manager_comment": null,
      "approved_at": "2026-06-10T14:30:00Z",
      "rejected_at": null
    },
    {
      "id": 41,
      "employee_username": "john_doe",
      "leave_type": "SICK",
      "start_date": "2026-06-08",
      "end_date": "2026-06-08",
      "number_of_days": 1,
      "reason": "Medical appointment",
      "status": "PENDING",
      "manager_comment": null,
      "approved_at": null,
      "rejected_at": null
    }
  ],
  "total": 2,
  "skip": 0,
  "limit": 10,
  "status_filter": null
}

```

Response Schema:

Field	Type	Description
items	array	Array of leave request objects
total	integer	Total number of records

Field	Type	Description
skip	integer	Number of records skipped
limit	integer	Limit applied
status_filter	string	Applied status filter (null if none)

Leave Request Object Schema:

Field	Type	Description
id	integer	Unique leave request ID
employee_username	string	Employee username
leave_type	string	Type of leave
start_date	date	Start date (YYYY-MM-DD)
end_date	date	End date (YYYY-MM-DD)
number_of_days	integer	Number of days
reason	string	Reason for leave
status	string	Current status (PENDING, APPROVED, REJECTED, CANCELLED)
manager_comment	string	Manager's comment (if any)
approved_at	datetime	Approval timestamp (if approved)
rejected_at	datetime	Rejection timestamp (if rejected)

Error Responses:

HTTP/1.1 400 Bad Request
Content-Type: application/json

```
{
  "detail": "Invalid status. Must be one of: PENDING, APPROVED, REJECTED, CANCELLED"
}
```

cURL Example - Get all approved leaves:

```
curl -X GET "http://localhost/leaves/history?status=APPROVED&limit=20" \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \
-H "Content-Type: application/json"
```

cURL Example - Get with pagination:

```
curl -X GET "http://localhost/leaves/history?skip=10&limit=5" \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \
-H "Content-Type: application/json"
```

GET /leaves/pending

Retrieve pending leave requests for the manager's team with advanced filtering.

Endpoint: GET http://localhost/leaves/pending

Access: Authenticated (Managers only)

Request Headers:

```
Authorization: Bearer <jwt_token>
Content-Type: application/json
```

Query Parameters:

Parameter	Type	Default	Description
skip	integer	0	Number of records to skip
limit	integer	10	Number of records to return
employee	string	None	Filter by employee username
status	string	PENDING	Filter by status
start_date	string	None	Filter by leave start date (YYYY-MM-DD)
end_date	string	None	Filter by leave end date (YYYY-MM-DD)

Success Response:

HTTP/1.1 200 OK

Content-Type: application/json

```

{
  "items": [
    {
      "id": 41,
      "employee_username": "john_doe",
      "leave_type": "SICK",
      "start_date": "2026-06-08",
      "end_date": "2026-06-08",
      "number_of_days": 1,
      "reason": "Medical appointment",
      "status": "PENDING",
      "manager_comment": null,
      "approved_at": null,
      "rejected_at": null
    },
    {
      "id": 40,
      "employee_username": "alice_johnson",
      "leave_type": "CASUAL",
      "start_date": "2026-06-20",
      "end_date": "2026-06-24",
      "number_of_days": 5,
      "reason": "Family vacation",
      "status": "PENDING",
      "manager_comment": null,
      "approved_at": null,
      "rejected_at": null
    }
  ],
  "total": 2,
  "skip": 0,
  "limit": 10,
  "filters": {
    "status": "PENDING",
    "employee": null,
    "start_date": null,
    "end_date": null
  }
}

```

Error Responses:

HTTP/1.1 403 Forbidden

Content-Type: application/json

```
{  
  "detail": "Managers only"  
}
```

HTTP/1.1 400 Bad Request

Content-Type: application/json

```
{  
  "detail": "Employee not in your team"  
}
```

cURL Example - Get pending leaves for specific employee:

```
curl -X GET "http://localhost/leaves/pending?employee=john_doe" \  
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
  -H "Content-Type: application/json"
```

cURL Example - Get pending leaves in date range:

```
curl -X GET "http://localhost/leaves/pending?start_date=2026-06-01&end_date=2026-06-30" \  
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
  -H "Content-Type: application/json"
```

PUT /leaves/{leave_id}/approve

Approve a pending leave request (Manager action).

Endpoint: PUT http://localhost/leaves/{leave_id}/approve

Access: Authenticated (Managers only - must be the reporting manager)

Request Headers:

Authorization: Bearer <jwt_token>

Content-Type: application/json

Path Parameters:

Parameter	Type	Required	Description
leave_id	integer	Yes	ID of the leave request

Query/Body Parameters: None

Success Response:

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "message": "Leave approved successfully",
  "leave_id": 41,
  "status": "APPROVED"
}
```

Response Schema:

Field	Type	Description
message	string	Success message
leave_id	integer	ID of the approved leave
status	string	New status (APPROVED)

Error Responses:

HTTP/1.1 403 Forbidden

Content-Type: application/json

```
{
  "detail": "Managers only"
}
```

HTTP/1.1 404 Not Found

Content-Type: application/json

```
{
  "detail": "Leave request not found"
}
```

HTTP/1.1 400 Bad Request

Content-Type: application/json

```
{
  "detail": "Only pending requests can be approved"
}
```

HTTP/1.1 400 Bad Request

Content-Type: application/json

```
{
  "detail": "Managers cannot approve their own leave"
}
```

HTTP/1.1 403 Forbidden

Content-Type: application/json

```
{
  "detail": "You cannot approve this employee's leave"
}
```

Processing Flow:

1. Verify current user is a manager
2. Check leave exists and is in PENDING status
3. Validate current user is the reporting manager of the employee
4. Deduct leave balance from employee's account
5. Update leave status to APPROVED
6. Publish event notification

cURL Example:

```
curl -X PUT "http://localhost/leaves/41/approve" \  
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
  -H "Content-Type: application/json"
```

PUT /leaves/{leave_id}/reject

Reject a pending leave request (Manager action).

Endpoint: PUT http://localhost/leaves/{leave_id}/reject

Access: Authenticated (Managers only - must be the reporting manager)

Request Headers:

Authorization: Bearer <jwt_token>
Content-Type: application/json

Path Parameters:

Parameter	Type	Required	Description
leave_id	integer	Yes	ID of the leave request

Query Parameters:

Parameter	Type	Required	Description
comment	string	Yes	Rejection reason/comment

Success Response:

HTTP/1.1 200 OK
Content-Type: application/json

```
{
  "message": "Leave rejected successfully",
  "leave_id": 41,
  "status": "REJECTED"
}
```

Response Schema:

Field	Type	Description
message	string	Success message
leave_id	integer	ID of the rejected leave
status	string	New status (REJECTED)

Error Responses:

HTTP/1.1 403 Forbidden
Content-Type: application/json

```
{
  "detail": "Managers only"
}
```

HTTP/1.1 404 Not Found
Content-Type: application/json

```
{
  "detail": "Leave request not found"
}
```

HTTP/1.1 400 Bad Request
Content-Type: application/json

```
{
  "detail": "Only pending requests can be rejected"
}
```

cURL Example:

```
curl -X PUT "http://localhost/leaves/41/reject?comment=Need+more+notice" \  
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." \  
  -H "Content-Type: application/json"
```

Error Handling

Standard Error Response Format

All error responses follow this format:

```
{  
  "detail": "Human-readable error message"  
}
```

Common Error Scenarios

Authentication Errors

Missing Authorization Header:

```
{  
  "detail": "Not authenticated"  
}
```

Expired Token:

```
{  
  "detail": "Invalid or expired token"  
}
```

Invalid Token:

```
{  
  "detail": "Invalid authentication credentials"  
}
```

Validation Errors

Invalid Request Body:

```
{  
  "detail": "Validation error details"  
}
```

Type validation example:

```
{  
  "detail": "value is not a valid integer"  
}
```

Service Errors

Service Unavailable:

```
{  
  "detail": "Employee Service temporarily unavailable"  
}
```

Database Errors:

```
{  
  "detail": "Database error"  
}
```

Status Codes

Standard HTTP Status Codes Used

Code	Meaning	When Used
200	OK	Successful GET, PUT, POST request with data response
201	Created	Resource successfully created
400	Bad Request	Invalid request parameters or validation failed
401	Unauthorized	Missing or invalid authentication token
403	Forbidden	User lacks permissions for the resource
404	Not Found	Resource does not exist
409	Conflict	Request conflicts with existing data (e.g., overlapping leave)
500	Internal Server Error	Unexpected server error
503	Service Unavailable	Dependent service is unavailable

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